BBC | Research & Development

Release Notes for HLG Format Conversion LUTs v1.8

Part of the HDR-TV series. Last updated September 2026.

These notes should be read in conjunction with the "Implementation Guidelines for HLG Format Conversion LUTs" and the "Guidance on Format Conversion in HLG Production".

Summary

To facilitate the introduction of ITU-R Recommendation BT.2100 Hybrid Log-Gamma (HLG) production, BBC R&D are licensing a package of look-up tables (LUTs) that implement a range of key format conversions.

This v1.8 release is for releasing under the Apache License version 2.0. File headers have been updated with the new licensing information. Whitespace in LUTs 8a and 8c has been adjusted to match the other LUT files. No technical changes to the LUTs have been made from the previous v1.7 release.

Starting with the v1.6 release, the LUTs are split into a series of folders targeting different applications; thereby helping to ensure that the optimum LUT is used in each case. They are further divided into primary and secondary conversions. The primary conversion LUTs are commonly used, whilst the secondary LUTs target a small number of specific applications. In addition, a number of LUTs have been moved to a "deprecated" folder. This is for conversions that are no longer recommended or considered useful, for example, the LUT variants with a "colour matrix compensation"; most LUT interpolators now allow users to select between BT.709 and BT.2100/BT.2020 colour matrices. See "Colour Matrices" in the "Implementation Guidelines for HLG Format Conversion LUTs" for more details.

The HLG outputs of LUTs 3, 4, 5, 6, 17, 18, 19 & 20 are clipped to the EBU R103 "preferred" signal range (-5% to + 105%), specified for SDR signals. The clipping level may be revised once more is known about the impact of the use of HDR sub-blacks[1] and super-whites[2] on video compression. The HLG outputs from LUTs 1, 2, 11 and 12 remain unclipped, to allow the highest dynamic range

[1] signals below black (BT.709 10-bit code value 64)

[2] signals above nominal peak-white (BT.709 10-bit code value 940)

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from PQ and S-Log3 signals to be transcoded to HLG. Similarly, the HLG input to LUT 7c and 7d is unclipped, to allow the entire HLG signal range to be transcoded to PQ.

LUT Header Metadata

From the v1.7 release, additional metadata is included in the LUT file headers of the Primary Live and Primary Post-Production conversion LUTs (see Table 2). The metadata, summarised in Table 1, is based on ITU-T H.273 CICP values. An *Input/Output Extended Range Processing* tag is also included. This indicates whether a LUT processes just the nominal signal range (*Extended Range Processing=0*), or an extended signal range, that includes the sub-blacks and super-whites (*Extended Range Processing=1*). For consistency with earlier LUT releases, the extended processing range is also described in the tables of Annex 1 as providing "Headroom".

The metadata can be used to,

- Automatically configure the interpolator controls for each LUT;
- Check that the input video signal format, for example carried in the SDI VPID signalling, matches that expected by the LUT;
- Add appropriate signalling (e.g. VPID) to correctly identify the output signal format to downstream devices.

As the metadata tags have not yet been standardised, they are included as comments preceded by the "#" character. The tags are considered "experimental" at this stage and may be ignored. There is no requirement for licensees to process the tags, but they are free to do so should they wish.

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Table 1 – LUT Header Metadata

Input Video Format	
Input Colour Primaries	CICP ITU-T H.273 Table 2
Input Matrix Coefficients	CICP ITU-T H.273 Table 4
Input Transfer Characteristics	CICP ITU-T H.273 Table 3
Input Extended Range Processing	Boolean
Output Video Format	
Output Colour Primaries	CICP ITU-T H.273 Table 2
Output Matrix Coefficients	CICP ITU-T H.273 Table 4
Output Transfer Characteristics	CICP ITU-T H.273 Table 3
Output Extended Range Processing	Boolean

LUT Details

The complete set of conversion LUTs provided to licensees as part of the v1.8 release is given below.

Table 2 – Licensed Conversion LUTs

BBC **LUT Number**	**Conversion**
Primary Live Production	
1e	BT.2100 PQ 1000 cd/m2 to BT.2100 HLG
2e	BT.2100 PQ 4000 cd/m2 to BT.2100 HLG
3c	BT.709 to BT.2100 HLG direct-mapping (display-light)
4-1a	BT.709 to BT.2100 HLG direct-mapping (scene-light)
5c	BT.709 to BT.2100 HLG up-mapping (display-light)
6-1a	BT.709 to BT.2100 HLG up-mapping (scene-light)
7c	BT.2100 HLG to BT.2100 PQ 1000 cd/m2
8c*	BT.2100 HLG to BT.709 CAM-based down-mapping (display-light)
9c	BT.2100 HLG to BT.709 down-mapping (display-light) with SDR super-whites
22c	BT.709 SDR camera shading LUT. Combines LUT 4 followed by LUT 9 into a single LUT
Primary Post-Production	
1a	BT.2100 PQ 1000 cd/m2 to BT.2100 HLG
2a	BT.2100 PQ 4000 cd/m2 to BT.2100 HLG
3a	BT.709 to BT.2100 HLG direct-mapping (display-light)
5a	BT.709 to BT.2100 HLG up-mapping (display-light)
7a	BT.2100 HLG to BT.2100 PQ 1000 cd/m2
8a	BT.2100 HLG to BT.709 CAM-based down-mapping (display-light)

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Secondary Conversions (Live and Post)	
1c	BT.2100 PQ 1000 cd/m2 to BT.2100 HLG
2c	BT.2100 PQ 4000 cd/m2 to BT.2100 HLG
10a	S-Log3 (100%) to BT.2100 HLG (Scene-Light)
11a	S-Log3 (200%) to BT.2100 HLG (Scene-Light)
14a	PQ P3D65 1000 cd/m^2 to BT.2100 HLG
15a	BT.2100 HLG to PQ P3D65 1000 cd/m^2
16a	BT.2100 HLG to X′Y′Z′ PQ at 108 cd/m^2
17a & c	BT.2020 to BT.2100 HLG direct-mapping (display-light)
18a & c	BT.2020 to BT.2100 HLG up-mapping (display-light)
19a & c	BT.2100 HLG traditional camera look-LUT – increases saturation and hue shift
20a & c	sRGB to BT.2100 HLG direct-mapping (display-light)
21c	BT.2100 HLG to SDR BT.2020 (Wide Colour Gamut) down-mapping (display-light)
DaVinci Resolve (Type III Headroom)	
1e	BT.2100 PQ 1000 cd/m2 to BT.2100 HLG
2e	BT.2100 PQ 4000 cd/m2 to BT.2100 HLG
3c	BT.709 to BT.2100 HLG direct-mapping (display-light)
5c	BT.709 to BT.2100 HLG up-mapping (display-light)
7c	BT.2100 HLG to BT.2100 PQ 1000 cd/m2
8c	BT.2100 HLG to BT.709 CAM-based down-mapping (display-light)
Test LUTs	
13a, b, c & d	Test LUTs
Deprecated Conversions	
All	Variants with colour matrix compensation
4-2	BT.709 to BT.2100 HLG direct-mapping (scene-light) (strict)
6-2	BT.709 to BT.2100 HLG up-mapping (scene-light) (strict)
12-1	BT.2100 HLG to BT.709 down-mapping (scene-light) (square root)
12-2	BT.2100 HLG to BT.709 down-mapping (scene-light) (strict)

* LUT8c should not be used in live production applications where round-trip performance is critical. See Annex 1.

The LUTs are described in detail in Annex 1 of this document. The annex tabulates the expected output values for SMPTE RP 219 BT.709 colour bars, BT.2111 HDR colour bars and the new EBU Tech 3373 HLG HDR colour bars. The EBU colour bars, with their embedded 75% BT.709 equivalent bars, are designed to be more robust to HDR/SDR format-conversion than the BT.2111 colour bars.

Changes between this v1.8 and the previous v1.7 release are highlighted.

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LUT Revision History

LUT Release	Date	Comments
1.0	22/12/17	• Initial release
1.1	27/03/18	• Adds S-Log3 to HLG conversion • Adds several Type III LUTs to pass sub-blacks and super-whites • Corrects a small black-level offset for PQ LUTs 1c, 1d and 2c, 2d, 7a and 7b
1.2	20/07/18	• SDR scene-light mapping to HLG (LUT 4) now maps 100% SDR to 75% HLG rather than 73%, to align with recommendations in ITU-R Report BT.2390. • Improves the "round-tripping" performance of the display-light SDR to HLG up-mapping (LUT 5) and HLG to SDR down-mapping (LUT 8). • Improves the colour-space conversion of the display-light HLG to SDR down-mapping (LUT 8) • Adds LUT 12 for scene-light HLG-to-SDR conversion, to mimic the SDR output of a camera CCU. • Adds LUTs 13a, 13b, 13c and 13d for testing LUT hardware. • Minor improvement to the accuracy of all full-range LUT conversions.
1.3	09/04/19	• Reduced hue-shifts in the HLG to SDR BT.709 display-light conversion; • Adds scene-light LUTs for different types of SDR cameras (square-root and strict BT.709 OETFs); • Reduces the scene-light inverse-tone mapping signal level to cope with a wider range of SDR signals in live production; • Adds conversions to/from PQ with P3D65 colorimetry; • Adds 65-cube LUTs for critical workflows; • Adds an HLG to HDR X′Y′Z′ PQ 110 cd/m^2 HDR projector LUT.
1.3.1	20/06/19	• Corrects a sub-black clipping problem in LUT 8c
1.3.2	17/10/19	• Corrects a scaling issue in the PQ to HLG Type III conversions, LUT s 7c & 7d
1.4	11/02/20	• LUT2, a new luminance-based tone-mapping curve is used for the PQ4000-to-HLG conversion, so that hue shifts on very bright saturated colours are eliminated. • LUT3, minor accuracy improvements. • LUT5, changes are made to the SDR to HLG up-mapping to minimise round-trip losses with LUTs 8 and 9.

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		• LUT8, the display-light HLG to SDR down-mapping is brighter than previous releases, to correctly-map the BT.709 equivalent colour bars within the EBU Tech 3373 HLG HDR colour bars to 75% BT.709. There are also colour-saturation improvements over the v1.3.1 release. • LUT9, a new HLG-to-SDR display-light LUT has been added (replacing the original LUT9, Dolby PRM4200/4220 HLG LUT – still available upon request). The new LUT minimises round-trip losses for directly-mapped SDR content (using LUT3) by shifting most of the HDR image-highlights to the SDR super-white region. • More 65-cubed versions of some of the key LUTs have been included for critical applications. • In addition, the HLG output is now clipped at EBU R103 "preferred" levels for LUTs 1, 2, 3, 4, 5, 6, 10 and 11
1.5	15/12/2020	• A new, easier to understand filename format. • LUTs 8 &9 - improved colour saturation in display-light HLG to SDR down-mapping. • LUTs17 & 18 - two new LUTs for BT.2020 SDR to BT.2100 HLG direct-mapping and up-mapping. • LUT19 - adjusts the hue and colour saturation of a BT2100 HLG camera output to match a traditional BT.709 camera; • LUT20 sRGB to HLG direct-mapping. • More "65-cubed" versions of some key LUTs have been included for critical applications.
1.6	21/12/2022	• Improves to the colour reproduction and detail of highlights in LUT 9 (HLG to SDR display-light down-mapping with "super-whites"). • Adds a new CAM (Colour Appearance Model) based down-mapping conversion, replacing the previous LUT8. • Adds a new HLG to SDR BT.2020 (wide colour gamut) display-light down-mapping conversion, LUT21. • Adds a new "combination" LUT for shading SDR BT.709 cameras in live "single stream" HDR production. The new LUT combines LUT4-1 (scene-light BT.709 to HLG) and LUT9 (display-light HLG to BT.709) into single conversion, thereby allowing camera shaders to directly view the SDR signal seen by audiences, LUT22. • Adds Type III LUTs that process sub-blacks and super-whites for use with DaVinci Resolve. Other software-based tools may be added in future releases, as vendors add the ability to process the sub-blacks and super-whites of "narrow range" signals.

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		• Splits the LUTs into a series of folders targeting different applications, helping to ensure that the correct LUT variant is used in each application. • Includes a folder for “deprecated” conversions that are no longer recommended or considered useful. • More “65-cubed” versions of some key LUTs have been included for critical applications.
1.7	07/03/2024	• Improves the reproduction of bright saturated colours in LUT 9 (HLG to SDR display-light down-mapping with "super-whites ") • A wider colour gamut SDR output when using the HLG to SDR BT.2020 display-light down-mapping conversion introduced in v1.6, LUT 21 • The addition of ITU-T H.273 CICP (Coding-independent Code Points) metadata tags in the LUT file header comments of certain LUTs, to facilitate interpolator configuration • Corrects release note errors in the “LUT 21 Details” table.
1.8	30/09/2026	• Updates LUT headers with new licence information following open source release. • Whitespace in files for LUT 8 (HLG to SDR CAM-based display-light down-mapping) has been adjusted to match other LUT files. • No technical changes to the LUTS.

Licensing

The LUTs are available under the Apache License, version 2.0, at https://github.com/bbc/bbc_hdr_television_production_luts

Please email transfer.rd@bbc.co.uk with any queries.

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Annex 1- Details of new or changed LUTs

Target Output Values

For the majority of LUTs, we provide tables of the ideal output values for defined input test signals. Please note that within the tables, the R′G′B′ output signals are normalised in the range 0.0 to 1.0, over the *nominal* signal range. They do not represent the output values from the look-up tables themselves, which will be different for each LUT variant. Furthermore, the values are derived from floating-point algorithmic implementations of the conversions. See the "Implementation Guidelines for HLG Format Conversion LUTs" for more details about the LUT variants and signal scaling. The tables of ideal values take no account of 3D-LUT interpolation errors, which will vary between implementations. Nor do they factor in the signal-clipping associated with Type I (narrow range mode) LUTs that do not process sub-blacks and super-whites. In addition, the 10-bit $Y'C'_BC'_R$ code values have been calculated from the floating-point R′G′B′ output values, rather than being rounded after 10-bit R′G′B′ hardware processing. The tables of ideal output values have been provided as an indication for implementers that they have chosen the correct LUT variant, for their particular LUT implementation. The tables are not intended to help debug hardware or software. Differences of +/- two 10-bit data levels should be expected between the tabulated 'ideal' values, and real-world 33-cubed and 65-cubed 3D-LUT implementations.

For the display-light and scene-light down-mapping LUTs (LUTs 8, 9, 12 and 21), the 3D-LUT interpolation errors tend to be higher than for the other LUTs. That is because the output signals extend to the extremities of the R′G′B′ signal range, where one or more of the index values (between which others are interpolated) may have been clipped to lie within the desired output signal range. So, for these LUTs we provide tables of the expected output values from a 33-cubed **tetrahedral** interpolator instead. Previous versions of these release notes used trilinear interpolation; however, tetrahedral interpolation is now more widely available and is always preferred. The errors are generally smaller with tetrahedral interpolation and the 65-cubed LUTs, but the worst-case errors may be similar.

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LUT 1: Graded Content – BT.2100 PQ 1000 cd/m^2 to BT.2100 HLG

Changes in v1.8: LUT headers have been updated with new licence information following open source release.

Primary conversion for live and post-production.

BT.2100 PQ signals are converted to BT.2100 HLG in the 1000 cd/m^2 "bridge" condition, so that 1000 cd/m^2 PQ maps to 100% HLG. Please refer to ITU-R report BT.2390 Section 7.2 ("Conversion concepts using a reference condition at 1000 cd/m^2"). With the Type III LUTs, PQ signals above 1000 cd/m^2 are mapped into the HLG "super-white" signal range up to 109% signal, equivalent to 1811 cd/m^2 (PQ).

The table illustrates the expected output signals from an ideal algorithmic converter, presented with a BT.2111 PQ colour bar input. It takes no account of the expected interpolation or rounding errors.

LUT 1: Graded Content - PQ 1000 cd/m2 to BT.2100 HLG						
Test Input: BT.2111 PQ Colour bars	**Ideal BT.2100 HLG Output**					
	Nominal Range [0.0 … 1.0]			**10-Bit SDI**		
	R'	**G'**	**B'**	**Y'**	**C'$_B$**	**C'$_R$**
100% White	1.090	1.090	1.090	1019	512	512
100% Yellow	1.090	1.090	0.000	962	24	551
100% Cyan	0.000	1.090	1.090	768	648	24
100% Green	0.000	1.090	0.000	711	160	63
100% Magenta	1.090	0.000	1.090	372	864	961
100% Red	1.090	0.000	0.000	315	376	1000
100% Blue	0.000	0.000	1.090	121	1000	473
58% White	0.7487	0.7487	0.7487	720	512	512
58% Yellow	0.7507	0.7507	0.0000	683	176	539
58% Cyan	0.0000	0.7586	0.7586	554	607	172
58% Green	0.0000	0.7613	0.0000	516	266	198
58% Magenta	0.7855	0.0000	0.7855	286	766	836
58% Red	0.7920	0.0000	0.0000	246	413	867
58% Blue	0.0000	0.0000	0.8394	108	888	482
40% Grey	0.4143	0.4143	0.4143	427	512	512
−7% Step	-0.0410	-0.0410	-0.0410	28	512	512
0% Step	0.0000	0.0000	0.0000	64	512	512
10% Step	0.0613	0.0613	0.0613	118	512	512
20% Step	0.1408	0.1408	0.1408	187	512	512
30% Step	0.2549	0.2549	0.2549	287	512	512
40% Step	0.4143	0.4143	0.4143	427	512	512
50% Step	0.6152	0.6152	0.6152	603	512	512
60% Step	0.7805	0.7805	0.7805	748	512	512
70% Step	0.9265	0.9265	0.9265	876	512	512

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80% Step	1.067	1.067	1.067	999	512	512
90% Step	1.090	1.090	1.090	1019	512	512
100% Step	1.090	1.090	1.090	1019	512	512
109% Step	1.090	1.090	1.090	1019	512	512
58% BT.709 Yellow	0.7425	0.7489	0.2897	695	307	526
58% BT.709 Cyan	0.5483	0.7434	0.7542	671	541	424
58% BT.709 Green	0.5247	0.7425	0.2717	640	328	431
58% BT.709 Magenta	0.7113	0.2797	0.7724	434	679	688
58% BT.709 Red	0.7076	0.2658	0.1290	391	395	715
58% BT.709 Blue	0.2298	0.1179	0.8123	229	809	537
0% Black	0.0000	0.0000	0.0000	64	512	512
−2% Black	-0.0125	-0.0125	-0.0125	53	512	512
+2% Black	0.0125	0.0125	0.0125	75	512	512
+4% Black	0.0245	0.0245	0.0245	85	512	512

LUT 1 Details

Variant	LUT Type	LUT Cube Size	SDI LUT Mode Narrow: 64 - 940 (10-bit) Full: 0 – 1023 (10-bit)	Input Range [Min Input .. Max input]	Output Range [0.0 .. 1.0]	BT.709 colour matrix comp.
1a	I	33	Narrow	Nominal	Nominal	No
1a65	I	65	Narrow	Nominal	Nominal	No
1b	I	33	Narrow	Nominal	Nominal	Yes
1c	II	33	Full	Full/Nominal	Headroom	No
1d	II	33	Full	Full/Nominal	Headroom	Yes
1e	III	33	Full	Headroom	Headroom	No
1e65	III	65	Full	Headroom	Headroom	No
1f	III	33	Full	Headroom	Headroom	Yes

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LUT 1 Header Metadata									
Variant	LUT Type	Input				Output			
		Colour Primaries	Matrix	Transfer Function	Extended Range Processing	Colour Primaries	Matrix	Transfer Function	Extended Range Processing
1a, 1a65, 1b	I	BT.2100 CICP 9	BT.2100 CICP 9	PQ CICP 16	0	BT.2100 CICP 9	BT.2100 CICP 9	HLG CICP 18	0
1c, 1d	II				0				1
1e, 1e65, 1f	III				1				1

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LUT 2: Graded Content – BT.2100 PQ 4000 cd/m^2 to BT.2100 HLG

Changes in v1.8: LUT headers have been updated with new licence information following open source release.

Primary conversion for live and post-production.

4000 cd/m^2 BT.2100 PQ signals are converted to BT.2100 HLG by first tone-mapping to the 1000 cd/m^2 "bridge" condition, and then converting to HLG. The tone-mapping is applied to the luminance component so that hue distortions are avoided; note that 4000 cd/m^2 PQ maps to 100% HLG. For more information, please refer to ITU-R report BT.2408 Section 6.4 ("Handling PQ signals with greater than 1000 cd/m^2 peak-luminance"). With Type III LUTs, PQ signals above 4000 cd/m^2 are mapped into the HLG "super-white" signal range.

The table below illustrates the output signals expected from an "ideal" algorithmic converter, presented with a BT.2111 PQ colour bar input. It takes no account of the expected interpolation or rounding errors.

LUT 2: PQ 4000 cd/m2 to BT.2100 HLG						
Test Input: BT.2111 PQ Bars	**Ideal BT.2100 HLG Output**					
	Nominal Range [0.0 … 1.0]			**10-Bit SDI**		
	R'	**G'**	**R'**	**Y'**	**C'$_B$**	**C'$_R$**
100% White	1.0380	1.0380	1.0380	973	512	512
100% Yellow	1.0042	0.9977	0.9844	939	505	515
100% Cyan	1.0506	0.9003	0.6824	876	396	587
100% Green	0.0000	1.0902	0.0000	711	160	63
100% Magenta	1.0902	0.0000	1.0902	372	864	961
100% Red	1.0902	0.0000	0.0000	315	376	1000
100% Blue	0.0000	0.0000	1.0902	121	1000	473
58% White	0.7487	0.7487	0.7487	720	512	512
58% Yellow	0.7507	0.7507	0.0000	683	176	539
58% Cyan	0.0000	0.7586	0.7586	554	607	172
58% Green	0.0000	0.7613	0.0000	516	266	198
58% Magenta	0.7855	0.0000	0.7855	286	766	836
58% Red	0.7920	0.0000	0.0000	246	413	867
58% Blue	0.0000	0.0000	0.8394	108	888	482
40% Grey	0.4143	0.4143	0.4143	427	512	512
−7% Step	-0.0410	-0.0410	-0.0410	28	512	512
0% Step	0.0000	0.0000	0.0000	64	512	512
10% Step	0.0613	0.0613	0.0613	118	512	512
20% Step	0.1408	0.1408	0.1408	187	512	512
30% Step	0.2549	0.2549	0.2549	287	512	512
40% Step	0.4143	0.4143	0.4143	427	512	512

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LUT 2: PQ 4000 cd/m2 to BT.2100 HLG						
Test Input: BT.2111 PQ Bars	**Ideal BT.2100 HLG Output**					
	Nominal Range [0.0 ... 1.0]			**10-Bit SDI**		
	R'	**G'**	**R'**	**Y'**	**C'$_B$**	**C'$_R$**
50% Step	0.6152	0.6152	0.6152	603	512	512
60% Step	0.7785	0.7785	0.7785	746	512	512
70% Step	0.8822	0.8822	0.8822	837	512	512
80% Step	0.9503	0.9503	0.9503	896	512	512
90% Step	0.9992	0.9992	0.9992	939	512	512
100% Step	1.0380	1.0380	1.0380	973	512	512
109% Step	1.0502	1.0502	1.0502	984	512	512
58% BT.709 Yellow	0.7425	0.7489	0.2897	695	307	526
58% BT.709 Cyan	0.5483	0.7434	0.7542	671	541	424
58% BT.709 Green	0.5247	0.7425	0.2717	640	328	431
58% BT.709	0.7113	0.2797	0.7724	434	679	688
58% BT.709 Red	0.7076	0.2658	0.1290	391	395	715
58% BT.709 Blue	0.2298	0.1179	0.8123	229	809	537
0% Black	0.0000	0.0000	0.0000	64	512	512
−2% Black	-0.0125	-0.0125	-0.0125	53	512	512
+2% Black	0.0125	0.0125	0.0125	75	512	512
+4% Black	0.0245	0.0245	0.0245	85	512	512

LUT 2 Details						
Variant	**LUT Type**	**LUT Cube Size**	**SDI LUT Mode Narrow: 64 - 940 (10-bit) Full: 0 – 1023 (10-bit)**	**Input Range [Min Input .. Max input]**	**Output Range [0.0 .. 1.0]**	**BT.709 colour matrix comp.**
2a	I	33	Narrow	Nominal	Nominal	No
2a65	I	65	Narrow	Nominal	Nominal	No
2b	I	33	Narrow	Nominal	Nominal	Yes
2c	II	33	Full	Full/Nominal	Headroom	No
2d	II	33	Full	Full/Nominal	Headroom	Yes
2e65	III	65	Full	Headroom	Headroom	No
2e	III	33	Full	Headroom	Headroom	No
2f	III	33	Full	Headroom	Headroom	Yes

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LUT 2 Header Metadata									
Variant	LUT Type	Input				Output			
		Colour Primaries	Matrix	Transfer Function	Extended Range Processing	Colour Primaries	Matrix	Transfer Function	Extended Range Processing
2a, 2a65, 2b	I	BT.2100 CICP 9	BT.2100 CICP 9	PQ CICP 16	0	BT.2100 CICP 9	BT.2100 CICP 9	HLG CICP 18	0
2c, 2d	II				0				1
2e, 2e65, 2f	III				1				1

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LUT 3: SDR Graphics - BT.709 to BT.2100 HLG, Direct-Mapping (Display-Light)

Changes in v1.8: LUT headers have been updated with new licence information following open source release.

Primary conversion for live and post-production.

BT.709 signals are directly-mapped into BT.2100 HLG at the BT.2408 signal levels using a display-light conversion. The "look" of the original BT.709 content is therefore preserved on conversion. A 100% SDR signal is mapped to 75% HLG ("HDR Reference White").

The table below illustrates the output signals expected from an "ideal" algorithmic converter, presented with a SMPTE RP 219 colour bar input. The expected interpolation errors have not been factored in.

LUT 3: BT.709 to BT.2100 HLG, Direct Mapping (Display-Light)						
Test Input: SMPTE 219 75% Colour bars	**Ideal BT.2100 HLG Output**					
	Nominal Range [0.0 … 1.0]			**10-Bit SDI**		
	R'	**G'**	**B'**	**Y'**	**C'_B**	**C'_R**
75% White	0.6145	0.6145	0.6145	602	512	512
75% Yellow	0.6055	0.6127	0.2073	578	331	523
75% Cyan	0.3930	0.6011	0.6131	543	543	418
75% Green	0.3702	0.5993	0.1914	515	358	424
75% Magenta	0.5363	0.1866	0.6060	330	656	654
75% Red	0.5233	0.1740	0.0848	292	428	672
75% Blue	0.1409	0.0722	0.6147	171	746	523
40% Grey	0.3107	0.3107	0.3107	336	512	512
15% Grey	0.1004	0.1004	0.1004	152	512	512
0% Black	0.0000	0.0000	0.0000	64	512	512
−2% Black	-0.0103	-0.0103	-0.0103	55	512	512
+2% Black	0.0103	0.0103	0.0103	73	512	512
100% White	0.7500	0.7500	0.7500	721	512	512

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LUT 3 Details						
Variant	**LUT Type**	**LUT Cube Size**	**SDI LUT Mode Narrow: 64 - 940 (10-bit) Full: 0 – 1023 (10-bit)**	**Input Range [Min Input .. Max input]**	**Output Range [0.0 .. 1.0]**	**BT.709 colour matrix comp.**
3a	I	33	Narrow	Nominal	Nominal	No
3a65	I	65	Narrow	Nominal	Nominal	No
3b	I	33	Narrow	Nominal	Nominal	Yes
3c	III	33	Full	Headroom	Headroom	No
3c65	III	65	Full	Headroom	Headroom	No
3d	III	33	Full	Headroom	Headroom	Yes

LUT 3 Header Metadata									
Variant	**LUT Type**	**Input**				**Output**			
		Colour Primaries	Matrix	Transfer Function	Extended Range Processing	Colour Primaries	Matrix	Transfer Function	Extended Range Processing
3a, 3a65, 3b	I	BT.709 CICP 1	BT.709 CICP 1	BT.709 CICP 1	0	BT.2100 CICP 9	BT.2100 CICP 9	HLG CICP 18	0
3c, 3c65, 3d	III				1				1

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LUT 4-1: SDR Compact Cameras – BT.709 to BT.2100 HLG, Direct Mapping (Scene-Light)

Changes in v1.8: LUT headers have been updated with new licence information following open source release.

Primary conversion for live production.

For use with SDR cameras that approximate the BT.709 OETF with a square-root

BT.709 camera signals are directly-mapped into BT.2100 HLG at the BT.2408 signal levels using a scene-light conversion. The “look” of the original BT.709 content is changed to match the subjective look of “native” BT.2100 HLG cameras. 100% SDR signal is directly-mapped to 75% HLG (“HDR Reference White”).

This LUT is intended to be used with SDR cameras with a limited dynamic-range, so no highlight “boost” is applied.

The table illustrates the output signals expected from an “ideal” converter, presented with a SMPTE RP 219 colour bar input. The expected interpolation errors have not been factored in.

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LUT 4-1: BT.709 to BT.2100 HLG, Direct Mapping (Scene-Light)						
Test Input: SMPTE 219 75% Colour bars	**Ideal BT.2100 HLG Output**					
	Nominal Range [0.0 ... 1.0]			**10-Bit SDI**		
	R'	**G'**	**B'**	**Y'**	**C'_B**	**C'_R**
75% White	0.6329	0.6329	0.6329	618	512	512
75% Yellow	0.6234	0.6304	0.2161	593	327	524
75% Cyan	0.4081	0.6175	0.6294	557	543	418
75% Green	0.3837	0.6149	0.1984	528	354	423
75% Magenta	0.5440	0.1897	0.6131	334	657	655
75% Red	0.5282	0.1758	0.0856	294	428	673
75% Blue	0.1391	0.0713	0.6092	170	744	523
40% Grey	0.3562	0.3562	0.3562	376	512	512
15% Grey	0.1333	0.1333	0.1333	181	512	512
0% Black	0.0000	0.0000	0.0000	64	512	512
−2% Black	-0.0183	-0.0183	-0.0183	48	512	512
+2% Black	0.0183	0.0183	0.0183	80	512	512
100% White	0.7500	0.7500	0.7500	721	512	512

LUT 4-1 Details						
Variant	**LUT Type**	**LUT Cube Size**	**SDI LUT Mode Narrow: 64 - 940 (10-bit) Full: 0 – 1023 (10-bit)**	**Input Range [Min Input .. Max input]**	**Output Range [0.0 .. 1.0]**	**BT.709 colour matrix comp.**
4-1a	III	33	Full	Headroom	Headroom	No
4-1a65	III	65	Full	Headroom	Headroom	No
4-1b	III	33	Full	Headroom	Headroom	Yes

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LUT 4-1 Header Metadata									
Variant	LUT Type	Input				Output			
		Colour Primaries	Matrix	Transfer Function	Extended Range Processing	Colour Primaries	Matrix	Transfer Function	Extended Range Processing
4-1a, 4-1a65, 4-1b	III	BT.709 CICP 1	BT.709 CICP 1	BT.709 CICP 1	1	BT.2100 CICP 9	BT.2100 CICP 9	HLG CICP 18	1

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LUT 4-2: SDR Compact Cameras – BT.709 to BT.2100 HLG, Direct Mapping (Scene-Light)

Changes in v1.8: LUT headers have been updated with new licence information following open source release.

Deprecated conversion – in practice, few SDR cameras implement a strict BT.709 OETF.

For use with SDR cameras that implement a strict BT.709 OETF

BT.709 camera signals are directly-mapped into BT.2100 HLG at the BT.2408 signal levels, using a scene-light conversion. The "look" of the original BT.709 content is changed to match the subjective look of "native" BT.2100 HLG cameras. 100% SDR signal is mapped to 75% HLG ("HDR Reference White").

This LUT is intended to be used with compact SDR cameras with a limited dynamic-range, so no highlight "boost" is applied.

NOTE: Unlike BT.2100 HLG, SDR cameras that implement a strict BT.709 OETF tend to crush detail in the shadows of a scene. Such detail will become more evident after applying this conversion. It is often better to adjust the SDR camera's native OETF to approximate a square-root (which provides a better match to HLG) and then use LUT 4-1 rather than this LUT.

The table below illustrates the output signals expected from an "ideal" converter, presented with a SMPTE RP 219 colour bar input. The expected interpolation errors have not been factored in.

LUT 4-2: BT.709 to BT.2100 HLG, Direct Mapping (Scene-Light)						
Test Input: SMPTE 219 75% Colour bars	**Ideal BT.2100 HLG Output**					
	Nominal Range [0.0 … 1.0]			**10-Bit SDI**		
	R'	**G'**	**B'**	**Y'**	**C'_B**	**C'_R**
75% White	0.6333	0.6333	0.6333	619	512	512
75% Yellow	0.6238	0.6309	0.2163	593	327	524
75% Cyan	0.4085	0.6180	0.6298	558	543	418
75% Green	0.3841	0.6153	0.1986	528	354	423
75% Magenta	0.5445	0.1899	0.6135	334	657	656
75% Red	0.5287	0.1760	0.0857	295	427	673
75% Blue	0.1393	0.0713	0.6096	170	745	523
40% Grey	0.3706	0.3706	0.3706	389	512	512
15% Grey	0.1711	0.1711	0.1711	214	512	512
0% Black	0.0000	0.0000	0.0000	64	512	512
−2% Black	-0.0602	-0.0602	-0.0602	11	512	512

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LUT 4-2: BT.709 to BT.2100 HLG, Direct Mapping (Scene-Light)						
Test Input: SMPTE 219 75% Colour bars	**Ideal BT.2100 HLG Output**					
	Nominal Range [0.0 … 1.0]			**10-Bit SDI**		
	R'	**G'**	**B'**	**Y'**	**C'_B**	**C'_R**
+2% Black	0.0602	0.0602	0.0602	117	512	512
100% White	0.7500	0.7500	0.7500	721	512	512

LUT 4-2 Details						
Variant	**LUT Type**	**LUT Cube Size**	**SDI LUT Mode Narrow: 64 - 940 (10-bit) Full: 0 – 1023 (10-bit)**	**Input Range [Min Input .. Max input]**	**Output Range [0.0 .. 1.0]**	**BT.709 colour matrix comp.**
4-2a	III	33	Full	Headroom	Headroom	No
4-2b	III	33	Full	Headroom	Headroom	Yes

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LUT 5: SDR Graded Content – BT.709 to BT.2100 HLG, Up-Mapping (Display-Light)

Changes in v1.8: LUT headers have been updated with new licence information following open source release.

Primary conversion for live and post-production.

BT.709 signals are up-mapped (inverse tone-mapped) to BT.2100 HLG using a display-light conversion. This LUT is designed to complement LUTs 8 and 9 (BT.2100 HLG to BT.709 down-mapping) so that losses associated with 'round-tripping' (i.e. SDR-to-HDR-to-SDR conversion) are minimised.

Whilst the ITU-R BT.2408 signal levels are taken into account, a modest boost is applied to the SDR highlights so that there is a closer match to natively-produced HDR content. The LUT does, however, attempt to preserve the artistic intent of the original BT.709 content. 100% SDR signal is up-mapped to 82% HLG.

The table below illustrates the output signals expected from an "ideal" converter, presented with a SMPTE RP 219 colour-bar input. The expected interpolation errors have not been factored in.

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LUT 5: BT.709 to BT.2100 HLG, Up-Mapping (Display-Light)						
Test Input: SMPTE 219 75% Colour bars	**Ideal BT.2100 HLG Output**					
	Nominal Range [0.0 … 1.0]			**10-Bit SDI**		
	R'	**G'**	**B'**	**Y'**	**C'_B**	**C'_R**
75% White	0.6145	0.6145	0.6145	602	512	512
75% Yellow	0.6055	0.6127	0.2073	578	331	523
75% Cyan	0.3930	0.6011	0.6131	543	543	418
75% Green	0.3702	0.5993	0.1914	515	358	424
75% Magenta	0.5363	0.1866	0.6060	330	656	654
75% Red	0.5233	0.1740	0.0848	292	428	672
75% Blue	0.1409	0.0722	0.6147	171	746	523
40% Grey	0.3107	0.3107	0.3107	336	512	512
15% Grey	0.1004	0.1004	0.1004	152	512	512
0% Black	0.0000	0.0000	0.0000	64	512	512
−2% Black	-0.0103	-0.0103	-0.0103	55	512	512
+2% Black	0.0103	0.0103	0.0103	73	512	512
100% White	0.8200	0.8200	0.8200	782	512	512

LUT 5 Details						
Variant	**LUT Type**	**LUT Cube Size**	**SDI LUT Mode Narrow: 64 - 940 (10-bit) Full: 0 – 1023 (10-bit)**	**Input Range [Min Input .. Max input]**	**Output Range [0.0 .. 1.0]**	**BT.709 colour matrix comp.**
5a	I	33	Narrow	Nominal	Nominal	No
5a65	I	65	Narrow	Nominal	Nominal	No
5b	I	33	Narrow	Nominal	Nominal	Yes
5c	III	33	Full	Headroom	Headroom	No
5c65	III	65	Full	Headroom	Headroom	No
5d	III	33	Full	Headroom	Headroom	Yes

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LUT 5 Header Metadata									
Variant	LUT Type	Input				Output			
		Colour Primaries	Matrix	Transfer Function	Extended Range Processing	Colour Primaries	Matrix	Transfer Function	Extended Range Processing
5a, 5a65,5b	I	BT.709 CICP 1	BT.709 CICP 1	BT.709 CICP 1	0	BT.2100 CICP 9	BT.2100 CICP 9	HLG CICP 18	0
5c, 5c65, 5d	III				1				1

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LUT 6-1: SDR High Quality Cameras – BT.709 to BT.2100 HLG, Up-Mapping (Scene-Light)

Changes in v1.8: LUT headers have been updated with new licence information following open source release.

Primary conversion for live production.

For use with SDR cameras that approximate the BT.709 OETF with a square root

BT.709 camera signals are up-mapped (inverse tone-mapped) to BT.2100 HLG, using a scene-light conversion. The "look" of the original BT.709 content is changed to match the "look" of native BT.2100 HLG cameras. Whilst the ITU-R BT.2408 signal levels are taken into account, a small boost is applied to the SDR highlights so that there's a close match to natively-produced HDR content. 100% SDR signal is up-mapped to 79% HLG. 105% SDR signals (EBU R.103 "preferred range" signals) are up-mapped to 83% HLG.

The table below illustrates the output signals expected from an "ideal" converter, presented with a SMPTE RP 219 colour bar input. The expected interpolation errors have not been factored in.

LUT 6-1: BT.709 to BT.2100 HLG, Up-Mapping (Scene-Light)						
Test Input: SMPTE 219 75% Colour bars	**Ideal BT.2100 HLG Output**					
	Nominal Range [0.0 … 1.0]			**10-Bit SDI**		
	R'	**G'**	**B'**	**Y'**	**C'_B**	**C'_R**
75% White	0.6329	0.6329	0.6329	618	512	512
75% Yellow	0.6234	0.6304	0.2161	593	327	524
75% Cyan	0.4081	0.6175	0.6294	557	543	418
75% Green	0.3837	0.6149	0.1984	528	354	423
75% Magenta	0.5440	0.1897	0.6131	334	657	655
75% Red	0.5282	0.1758	0.0856	294	428	673
75% Blue	0.1391	0.0713	0.6092	170	744	523
40% Grey	0.3562	0.3562	0.3562	376	512	512
15% Grey	0.1333	0.1333	0.1333	181	512	512
0% Black	0.0000	0.0000	0.0000	64	512	512
−2% Black	-0.0183	-0.0183	-0.0183	48	512	512
+2% Black	0.0183	0.0183	0.0183	80	512	512
100% White	0.7900	0.7900	0.7900	756	512	512

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LUT 6-1 Details						
Variant	**LUT Type**	**LUT Cube Size**	**SDI LUT Mode Narrow: 64 - 940 (10-bit) Full: 0 – 1023 (10-bit)**	**Input Range [Min Input .. Max input]**	**Output Range [0.0 .. 1.0]**	**BT.709 colour matrix comp.**
6-1a	III	33	Full	Headroom	Headroom	No
6-1a65	III	33	Full	Headroom	Headroom	No
6-1b	III	33	Full	Headroom	Headroom	Yes

LUT 6-1 Header Metadata									
Variant	**LUT Type**	**Input**				**Output**			
		Colour Primaries	Matrix	Transfer Function	Extended Range Processing	Colour Primaries	Matrix	Transfer Function	Extended Range Processing
6-1a, 6-1a65, 6-1b	III	BT.709 CICP 1	BT.709 CICP 1	BT.709 CICP 1	1	BT.2100 CICP 9	BT.2100 CICP 9	HLG CICP 18	1

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LUT 6-2: SDR High Quality Cameras – BT.709 to BT.2100 HLG, Up-Mapping (Scene-Light)

Changes in v1.8: LUT headers have been updated with new licence information following open source release.

Deprecated conversion – in practice, few SDR cameras implement a strict BT.709 OETF.

For use with SDR cameras that implement a strict BT.709 OETF

BT.709 camera signals are up-mapped (inverse tone-mapped) to BT.2100 HLG using a scene-light conversion. The "look" of the original BT.709 content is changed, so that the native look of BT.2100 HLG cameras is achieved. Whilst the ITU-R BT.2408 signal levels are taken into account, a small boost is applied to the SDR highlights so that there is a closer match to natively-produced HDR content. A 100% SDR signal is up-mapped to 79% HLG. 105% SDR signals (EBU R.103 "preferred range" signals) are up-mapped to 83% HLG.

NOTE: Unlike BT.2100 HLG, SDR cameras that implement a strict BT.709 OETF tend to crush detail in the shadows of a scene. Such detail will become more visible after applying this conversion. It is often better to adjust the SDR camera's native OETF to approximate a square-root (which provides a better match to HLG) and then use LUT 6-1.

The table below illustrates the output signals expected from an "ideal" converter, presented with a SMPTE RP 219 colour bar input. The expected interpolation errors have not been factored in.

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LUT 6-2: BT.709 to BT.2100 HLG, Up-Mapping (Scene-Light)						
Test Input: SMPTE 219 75% Colour bars	**Ideal BT.2100 HLG Output**					
	Nominal Range [0.0 ... 1.0]			**10-Bit SDI**		
	R'	**G'**	**B'**	**Y'**	**C'_B**	**C'_R**
75% White	0.6333	0.6333	0.6333	619	512	512
75% Yellow	0.6238	0.6309	0.2163	593	327	524
75% Cyan	0.4085	0.6180	0.6298	558	543	418
75% Green	0.3841	0.6153	0.1986	528	354	423
75% Magenta	0.5445	0.1899	0.6135	334	657	656
75% Red	0.5287	0.1760	0.0857	295	427	673
75% Blue	0.1393	0.0713	0.6096	170	745	523
40% Grey	0.3706	0.3706	0.3706	389	512	512
15% Grey	0.1711	0.1711	0.1711	214	512	512
0% Black	0.0000	0.0000	0.0000	64	512	512
−2% Black	-0.0602	-0.0602	-0.0602	11	512	512
+2% Black	0.0602	0.0602	0.0602	117	512	512
100% White	0.7900	0.7900	0.7900	756	512	512

LUT 6-2 Details						
Variant	**LUT Type**	**LUT Cube Size**	**SDI LUT Mode Narrow: 64 - 940 (10-bit) Full: 0 – 1023 (10-bit)**	**Input Range [Min Input .. Max input]**	**Output Range [0.0 .. 1.0]**	**BT.709 colour matrix comp.**
6-2a	III	33	Full	Headroom	Headroom	No
6-2b	III	33	Full	Headroom	Headroom	Yes

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LUT 7 – BT.2100 HLG to BT.2100 PQ 1000 cd/m^2

Changes in v1.8: LUT headers have been updated with new licence information following open source release.

Primary conversion for live and post-production.

BT.2100 HLG signals are converted to BT.2100 PQ at the 1000 cd/m^2 "bridge" condition, so that 100% HLG maps to 1000 cd/m^2 PQ. See ITU-R report BT.2390 Section 7.2 ("Conversion concepts using a reference condition at 1000 cd/m^2"). For Type III LUTs, HLG signals above 100% are mapped to PQ signals greater than 1000 cd/m^2, up to a maximum of 1810 cd/m^2 (corresponding to 109% HLG 'super-white', i.e. 10-bit code value 1019).

The table illustrates the output signals expected from an "ideal" converter, presented with an EBU Tech 3373 HLG colour bar input. The expected interpolation errors have not been factored in.

	LUT 7 - HLG to PQ 1000 cd/m^2						
	Test Input: BT.2100 HLG EBU Tech 3373	Ideal BT.2100 PQ (Narrow Range) Output					
		Nominal Range [0.0 ... 1.0]			10-Bit SDI		
		R'	G'	B'	Y'	C'$_B$	C'$_R$
100% HLG	100% White	0.7518	0.7518	0.7518	723	512	512
	100% Yellow	0.7505	0.7505	0.0000	682	176	539
	100% Cyan	0.0000	0.7452	0.7452	545	605	178
	100% Green	0.0000	0.7434	0.0000	506	272	206
	100% Magenta	0.7271	0.0000	0.7271	269	747	812
	100% Red	0.7227	0.0000	0.0000	230	422	836
	100% Blue	0.0000	0.0000	0.6904	100	821	487
75% HLG	75% White	0.5808	0.5808	0.5808	573	512	512
	75% Yellow	0.5795	0.5795	0.0000	542	252	533
	75% Cyan	0.0000	0.5744	0.5744	435	584	255
	75% Green	0.0000	0.5727	0.0000	404	327	276
	75% Magenta	0.5572	0.0000	0.5572	221	692	742
	75% Red	0.5530	0.0000	0.0000	191	443	760
	75% Blue	0.0000	0.0000	0.5226	91	746	493
	40% Grey	0.3916	0.3916	0.3916	407	512	512
Display Light Equivalent BT.709	DL White	0.4994	0.4994	0.4994	502	512	512
	DL Yellow	0.4939	0.4975	0.2959	489	422	518
	DL Cyan	0.4015	0.4892	0.4944	473	525	473
	DL Green	0.3890	0.4863	0.2797	457	432	476

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	LUT 7 - HLG to PQ 1000 cd/m²						
	Test Input: BT.2100 HLG EBU Tech 3373	Ideal BT.2100 PQ (Narrow Range) Output					
		Nominal Range [0.0 ... 1.0]			10-Bit SDI		
		R'	G'	B'	Y'	C'_B	C'_R
	DL Magenta	0.4424	0.2621	0.4721	346	584	585
	DL Red	0.4317	0.2480	0.1589	319	449	597
	DL Blue	0.2058	0.1325	0.4501	213	645	533
Scene Light Equivalent BT.709	SL White	0.5097	0.5097	0.5097	510	512	512
	SL Yellow	0.5039	0.5071	0.3031	497	421	518
	SL Cyan	0.4102	0.4980	0.5033	480	525	472
	SL Green	0.3969	0.4951	0.2862	464	431	476
	SL Magenta	0.4463	0.2653	0.4758	349	584	586
	SL Red	0.4343	0.2502	0.1606	321	449	598
	SL Blue	0.2045	0.1308	0.4477	212	645	534
Near Black Test Signal	-4%	-0.0610	-0.0610	-0.0610	11	512	512
	-2%	-0.0287	-0.0287	-0.0287	39	512	512
	-1%	-0.0123	-0.0123	-0.0123	53	512	512
	0%	0.0000	0.0000	0.0000	64	512	512
	1%	0.0123	0.0123	0.0123	75	512	512
	2%	0.0287	0.0287	0.0287	89	512	512
	4%	0.0610	0.0610	0.0610	117	512	512

LUT 7 Details						
Variant	**LUT Type**	**LUT Cube Size**	**SDI LUT Mode Narrow: 64 - 940 (10-bit) Full: 0 – 1023 (10-bit)**	**Input Range [Min Input .. Max input]**	**Output Range [0.0 .. 1.0]**	**BT.709 colour matrix comp.**
7a	I	33	Narrow	Nominal	Nominal	No
7a65	I	65	Narrow	Nominal	Nominal	No
7b	I	33	Narrow	Nominal	Nominal	Yes
7c	III	33	Full	Headroom	Headroom	No
7c65	III	65	Full	Headroom	Headroom	No
7d	III	33	Full	Headroom	Headroom	Yes

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LUT 7 Header Metadata									
Variant	LUT Type	Input				Output			
		Colour Primaries	Matrix	Transfer Function	Extended Range Processing	Colour Primaries	Matrix	Transfer Function	Extended Range Processing
7a, 7a65, 7b	I	BT.2100 CICP 9	BT.2100 CICP 9	HLG CICP 18	0	BT.2100 CICP 9	BT.2100 CICP 9	PQ CICP 16	0
7c, 7c65, 7d	III				1				1

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LUT 8: HDR Graded Content - BT.2100 HLG to BT.709 CAM-Based Down-Mapping (Display-Light)

Changes in v1.8: LUT headers have been updated with new licence information following open source release. Whitespace has been adjusted to match other LUT files.

Primary conversion for post-production.

BT.2100 HLG signals are down-mapped (tone-mapped) to BT.709 using a model of the human visual system (CAM). The LUT attempts to preserve the artistic intent of the original HDR content. As the LUT is based on a CAM, the appearance of colours, mid-tones and lowlights after conversion are significantly improved. This disadvantage if this approach is that it does not cascade as well as LUT9c, with the SDR to HLG direct-mapping and up-mapping LUTs (3 & 5). LUT8 is, therefore, not currently recommended for live productions, in which it is important to minimise 'round-tripping' losses (i.e. SDR-to-HDR-to-SDR conversion). It may, however, still be used for live production where that is not the case.

"HDR Reference White" (75% HLG) is tone-mapped to 88% BT.709, facilitating signal headroom for compressed highlights. 100% HLG signal is tone-mapped to 100% BT.709. With Type III LUTs that process sub-blacks and super-whites, the HLG super-white signals are tone-mapped to SDR super-white signals. The LUT outputs are clipped to EBU R.103 signal levels.

The 3D-LUT interpolation errors are greatest for the HDR to SDR down-mapping LUTs, as the output signals reach the extremities of the R′G′B′ signal range. For that reason, we provide the expected output values from a 33-cubed tetrahedral interpolator for the EBU Tech 3373 HLG Colour Bars, rather than the ideal values. Errors should be smaller with the 65-cubed LUTs and real pictures.

NOTE: Neither the EBU Tech 3373 HLG colour bars nor the ITU-R BT.2111 HLG colour bars will look like standard BT.709 colour bars after conversion, as the BT.2100 colour-primaries are quite different to those of BT.709. Furthermore, the converted colour bars will not look like BT.2111 colour bars displayed on a current reference display as their colour-primaries tend to be closer to DCI-P3 than BT.2100. Most noticeable might be the BT.2100 green, which is much more cyan in colour than the DCI-P3 and BT.709 greens.

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	LUT 8 - BT.2100 HLG to BT.709, Down-Mapping (Display-Light)						
	Test Input: BT.2100 HLG EBU Tech 3373	Expected BT.709 Output with tetrahedral interpolation					
		Nominal Range [0.0 ... 1.0]			10-Bit SDI		
		R'	G'	B'	Y'	C'_B	C'_R
100% HLG	100% White	0.9999	0.9999	0.9999	939	512	512
	100% Yellow	0.9959	0.9827	-0.0502	861	47	560
	100% Cyan	-0.0502	0.9893	0.9595	735	605	47
	100% Green	0.4066	0.9668	-0.0502	742	113	302
	100% Magenta	0.9884	-0.0502	0.8142	268	792	941
	100% Red	0.9560	-0.0502	-0.0426	207	412	962
	100% Blue	0.1519	-0.0502	0.9730	122	949	560
75% HLG	75% White	0.8797	0.8797	0.8797	834	512	512
	75% Yellow	0.8748	0.8617	-0.0502	763	102	555
	75% Cyan	-0.0502	0.8491	0.8240	638	593	110
	75% Green	0.2668	0.8355	-0.0502	633	173	293
	75% Magenta	0.8488	-0.0502	0.7055	235	758	883
	75% Red	0.8219	-0.0502	0.0937	191	486	896
	75% Blue	0.1157	-0.0502	0.7857	103	869	551
	40% Grey	0.4914	0.4914	0.4914	494	512	512
Display Light Equivalent BT.709	DL White	0.7501	0.7501	0.7501	721	512	512
	DL Yellow	0.7488	0.7504	-0.0502	670	153	544
	DL Cyan	0.0232	0.7498	0.7484	585	585	186
	DL Green	0.1439	0.7473	-0.0107	558	234	272
	DL Magenta	0.7347	-0.0354	0.7287	224	775	825
	DL Red	0.7174	0.0365	0.0120	221	431	818
	DL Blue	-0.0314	-0.0378	0.7038	78	843	484
Scene Light Equivalent BT.709	SL White	0.7736	0.7736	0.7736	741	512	512
	SL Yellow	0.7714	0.7714	-0.0502	687	143	545
	SL Cyan	0.0304	0.7696	0.7689	600	587	180
	SL Green	0.1400	0.7705	-0.0151	571	224	261
	SL Magenta	0.7452	-0.0216	0.7384	235	773	824
	SL Red	0.7250	0.0409	0.0112	225	428	819
	SL Blue	-0.0165	-0.0378	0.6967	81	838	491
Near Black Test Signal	-4%	-0.0475	-0.0475	-0.0475	22	512	512
	-2%	-0.0237	-0.0237	-0.0237	43	512	512
	-1%	-0.0119	-0.0119	-0.0119	53	512	512
	0%	0.0000	0.0000	0.0000	64	512	512
	1%	0.0119	0.0119	0.0119	74	512	512
	2%	0.0237	0.0237	0.0237	84	512	512
	4%	0.0475	0.0475	0.0475	105	512	512

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LUT 8 Details						
Variant	**LUT Type**	**LUT Cube Size**	**SDI LUT Mode Narrow: 64 - 940 (10-bit) Full: 0 – 1023 (10-bit)**	**Input Range [Min Input .. Max input]**	**Output Range [0.0 .. 1.0]**	**BT.709 colour matrix comp.**
8a	I	33	Narrow	Nominal	Nominal	No
8a65	I	65	Narrow	Nominal	Nominal	No
8c	III	33	Full	Headroom	Headroom	No
8c65	III	65	Full	Headroom	Headroom	No

LUT 8 Header Metadata										
Variant	**LUT Type**	**Input**				**Output**				
		Colour Primaries	Matrix	Transfer Function	Extended Range Processing	Colour Primaries	Matrix	Transfer Function	Extended Range Processing
8a, 8a65	I	BT.2100 CICP 9	BT.2100 CICP 9	HLG CICP 18	0	BT.709 CICP 1	BT.709 CICP 1	BT.709 CICP 1	0
8c, 8c65	III				1				1

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LUT 9: HDR-graded content and live programmes with graphics - BT.2100 HLG to BT.709, down-mapping (display-light) using SDR “super-whites”

Changes in v1.8: LUT headers have been updated with new licence information following open source release.

Primary conversion for live production.

BT.2100 HLG signals are down-mapped (tone-mapped) to BT.709 using a display-light conversion. Colours and the appearance of mid-tones and lowlights are maintained after conversion. The LUT, therefore, attempts to preserve the artistic intent of the original HLG content.

“HDR Reference White” (75% HLG) is mapped to 95% BT.709 and HDR highlights extend into the SDR super-white region; for that reason, only “Full-Range Mode” Type III LUTs are made available. 100% HLG signal is mapped to the EBU R103 preferred maximum of 105% BT.709. HLG signals above 100% are clipped.

This LUT is recommended for use in scenarios where minimising SDR>HDR>SDR “round-trip” losses is of the utmost importance. Minimal round-trip losses are achieved when used with the display-light up-mapping LUT5. A small loss of 5% to the SDR nominal peak white is incurred in combination with the SDR direct-mapping LUT3. The low round-trip losses are achieved by exploiting the SDR super-white signal range. Where there is a risk of the SDR super-white signal range being clipped, LUT8 may give better results at the expense of higher “round-trip” losses.

The 3D-LUT interpolation errors are greatest for the HDR to SDR down-mapping LUTs, as the output signals reach the extremities of the R′G′B′ signal range. For that reason, we provide the expected output values from a 33-cubed tetrahedral interpolator for the EBU Tech 3373 HLG Colour Bars, rather than the ideal values. Errors should be smaller with the 65-cubed LUTs and real pictures.

NOTE: Neither the EBU Tech 3373 HLG colour bars nor the ITU-R BT.2111 HLG colour bars will look like standard BT.709 colour bars after conversion, as the BT.2100 colour primaries are quite different from those of BT.709. Furthermore, the converted colour bars will not look like BT.2111 colour bars displayed on a current reference display, as their colour primaries tend to be closer to DCI-P3 than BT.2100. Of these, the most noticeable might be the BT.2100 green. This contains more cyan content than the DCI-P3 and BT.709 greens.

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	LUT 9 - BT.2100 HLG to BT.709, Down-Mapping (Display-Light)						
	Test Input: BT.2100 HLG EBU Tech 3373	Expected BT.709 Output with tetrahedral interpolation					
		Nominal Range [0.0 ... 1.0]			10-Bit SDI		
		R'	G'	B'	Y'	C'_B	C'_R
100% HLG	100% White	1.0495	1.0490	1.0490	984	512	512
	100% Yellow	1.0174	1.0230	0.4285	920	248	532
	100% Cyan	-0.0502	1.0014	0.9685	744	604	44
	100% Green	-0.0501	0.9461	0.3536	668	348	92
	100% Magenta	1.0502	0.2826	0.9145	496	716	828
	100% Red	1.0150	-0.0502	0.2154	236	520	980
	100% Blue	0.0711	0.1615	1.0502	244	920	436
75% HLG	75% White	0.9484	0.9483	0.9482	896	512	512
	75% Yellow	0.9409	0.9341	-0.0502	820	72	556
	75% Cyan	-0.0502	0.9083	0.8785	680	596	84
	75% Green	-0.0502	0.8670	0.3248	620	364	124
	75% Magenta	0.9974	-0.0502	0.8639	272	812	944
	75% Red	0.9502	-0.0502	0.2024	224	524	948
	75% Blue	0.0488	-0.0502	0.9510	100	952	516
	40% Grey	0.4966	0.4966	0.4966	500	512	512
Display Light Equivalent BT.709	DL White	0.7489	0.7489	0.7489	720	512	512
	DL Yellow	0.7476	0.7498	0.0555	676	200	540
	DL Cyan	0.0313	0.7477	0.7465	584	584	192
	DL Green	0.1003	0.7455	0.0468	552	264	252
	DL Magenta	0.7486	0.0112	0.7482	256	768	812
	DL Red	0.7442	0.0381	0.0345	228	436	828
	DL Blue	0.0503	0.0017	0.7448	120	840	504
Scene Light Equivalent BT.709	SL White	0.7773	0.7773	0.7773	744	512	512
	SL Yellow	0.7754	0.7762	0.0543	700	188	540
	SL Cyan	0.0360	0.7719	0.7715	604	588	184
	SL Green	0.1059	0.7684	0.1131	572	288	244
	SL Magenta	0.7590	0.0037	0.7580	256	772	820
	SL Red	0.7524	0.0403	0.0308	232	436	832
	SL Blue	0.0532	0.0054	0.7375	124	836	504
Near Black Test Signal	-4%	-0.0396	-0.0396	-0.0396	28	512	512
	-2%	-0.0198	-0.0198	-0.0198	48	512	512
	-1%	-0.0099	-0.0099	-0.0099	56	512	512
	0%	0.0000	0.0000	0.0000	64	512	512
	1%	0.0099	0.0099	0.0099	72	512	512
	2%	0.0197	0.0197	0.0197	80	512	512
	4%	0.0394	0.0394	0.0394	100	512	512

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LUT 9 Details						
Variant	**LUT Type**	**LUT Cube Size**	**SDI LUT Mode Narrow: 64 - 940 (10-bit) Full: 0 – 1023 (10-bit)**	**Input Range [Min Input .. Max input]**	**Output Range [0.0 .. 1.0]**	**BT.709 colour matrix comp.**
9c	III	33	Full	Headroom	Headroom	No
9c65	III	65	Full	Headroom	Headroom	No
9d	III	33	Full	Headroom	Headroom	Yes

LUT 9 Header Metadata									
Variant	**LUT Type**	**Input**				**Output**			
		Colour Primaries	Matrix	Transfer Function	Extended Range Processing	Colour Primaries	Matrix	Transfer Function	Extended Range Processing
9c 9c65 9d	III	BT.2100 CICP 9	BT.2100 CICP 9	HLG CICP 18	1	BT.709 CICP 1	BT.709 CICP 1	BT.709 CICP 1	1

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LUT 10: Sony S-Log3 cameras in post-production - S-Log3 (100%) to BT.2100 HLG (Scene-Light)

Changes in v1.8: LUT headers have been updated with new licence information following open source release.

Secondary conversion - most S-Log3 cameras can also capture BT.2100 HLG.

S-Log3 (BT.2020 colour) signals produced using the Sony's "100%" workflow (i.e. 100% IRE input equals 90% reflectance) are converted to BT.2100 HLG with ITU-R BT.2408 signal levels. A 100% IRE input signal (90% reflectance) thus maps to 73% HLG.

A scene-light conversion is used, so that the converted S-Log3 signal is a close subjective-match to the "look" of BT.2100 HLG cameras.

LUT 10: S-Log3 (100%) to BT.2100 HLG (Scene-Light)							
Test Input: S-Log3		Ideal BT.2100 HLG Output					
		Nominal Range [0.0 ... 1.0]			10-Bit SDI		
Reflectance	S-Log3 Signal	R'	G'	B'	Y'	C'_B	C'_R
0%	9%	0.0000	0.0000	0.0000	64	512	512
90%	58%	0.7292	0.7292	0.7292	703	512	512
100%	60%	0.7500	0.7500	0.7500	721	512	512

LUT 10 Details						
Variant	LUT Type	LUT Cube Size	SDI LUT Mode Narrow: 64 - 940 (10-bit) Full: 0 – 1023 (10-bit)	Input Range [Min Input .. Max input]	Output Range [0.0 .. 1.0]	BT.709 colour matrix comp.
10a	II	33	Full	Footroom	Headroom	No
10b	II	33	Full	Footroom	Headroom	Yes

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LUT 11: Sony S-Log3 cameras in "SR Live" workflows – S-Log3 (200%) to BT.2100 HLG (Scene-Light)

Changes in v1.8: LUT headers have been updated with new licence information following open source release.

Secondary conversion - most S-Log3 cameras can also capture BT.2100 HLG.

S-Log3 (BT.2020 colour) signals produced using Sony's "SR Live" "200%" workflow (i.e. 200% IRE input equals 90% reflectance) are converted to BT.2100 HLG with ITU-R BT.2408 signal levels. A 200% IRE input signal (90% reflectance) thus maps to 73% HLG. 200% workflows are commonly-encountered whenever the S-Log3 curve has been used in live productions.

A scene-light conversion is used, so that the converted S-Log3 signal matches the "look" of BT.2100 HLG cameras.

LUT 11: S-Log3 SR Live (200%) to BT.2100 HLG (Scene-Light)							
Test Input: S-Log3 SR Live		Ideal BT.2100 HLG Output					
		Nominal Range [0.0 ... 1.0]			10-Bit SDI		
Reflectance	S-Log3 Signal	R'	G'	B'	Y'	C'_B	C'_R
0%	9%	0.0000	0.0000	0.0000	64	512	512
90%	66%	0.7292	0.7292	0.7292	703	512	512
100%	67%	0.7500	0.7500	0.7500	721	512	512

LUT 11 Details						
Variant	LUT Type	LUT Cube Size	SDI LUT Mode Narrow: 64 - 940 (10-bit) Full: 0 – 1023 (10-bit)	Input Range [Min Input .. Max input]	Output Range [0.0 .. 1.0]	BT.709 colour matrix comp.
11a	II	33	Full	Footroom	Headroom	No
11b	II	33	Full	Footroom	Headroom	Yes

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LUT 12-1: "Clean" (i.e. without graphics) feeds to match SDR BT.709 cameras – BT.2100 HLG to BT.709, down-mapping (scene-light)

Changes in v1.8: LUT headers have been updated with new licence information following open source release.

Deprecated conversion – few workflows require this conversion. Better results obtained with LUT 9.

For matching downstream SDR cameras that approximate the BT.709 OETF with a square-root function

BT.2100 HLG signals are down-mapped (tone-mapped) to BT.709, using scene-light conversion. The "look" of the original BT.2100 HLG content is changed to match "native" BT.709 cameras with a square-root approximation of the BT.709 OETF.

This LUT is the *exact inverse* of LUT 6-1 (BT.709 to BT.2100 up-mapping, scene-light), so that the losses associated with 'round-tripping' (i.e. SDR-to-HDR-to-SDR conversion) are minimised.

"HDR Reference White" (75% HLG) is tone-mapped to 95% BT.709, allowing for some soft-clipping of highlights. A 79% HLG signal is tone-mapped to 100% BT.709. With Type III LUTs that process sub-blacks and super-whites, the LUT outputs are clipped to EBU R.103 signal levels.

The 3D-LUT interpolation errors are greatest for the HDR to SDR down-mapping LUTs, as the output signals reach the extremities of the R′G′B′ signal range. For that reason, we provide the expected output values from a 33-cubed tetrahedral interpolator for the EBU Tech 3373 HLG Colour Bars, rather than the ideal values. Errors should be smaller with the 65-cubed LUTs and real pictures.

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	LUT 12-1 - BT.2100 HLG to BT.709, Down-Mapping (Scene-Light)						
	Test Input: BT.2100 HLG EBU Tech 3373	Expected BT.709 Output with tetrahedral interpolation					
		Nominal Range [0.0 ... 1.0]			10-Bit SDI		
		R'	G'	B'	Y'	C'_B	C'_R
100% HLG	100% White	1.0502	1.0502	1.0502	984	512	512
	100% Yellow	1.0502	1.0502	-0.0502	914	18	557
	100% Cyan	-0.0502	1.0502	1.0502	779	624	18
	100% Green	-0.0502	1.0502	-0.0502	709	131	64
	100% Magenta	1.0502	-0.0502	1.0502	294	892	959
	100% Red	1.0502	-0.0502	-0.0502	224	399	1005
	100% Blue	-0.0502	-0.0502	1.0502	89	1005	466
75% HLG	75% White	0.9459	0.9459	0.9459	892	512	512
	75% Yellow	0.9917	0.9614	-0.0502	847	55	567
	75% Cyan	-0.0502	1.0310	1.0006	763	609	28
	75% Green	-0.0502	1.0328	-0.0502	698	137	71
	75% Magenta	1.0502	-0.0502	1.0264	293	881	960
	75% Red	1.0502	-0.0502	-0.0502	224	399	1005
	75% Blue	-0.0502	-0.0502	1.0300	88	995	467
	40% Grey	0.4479	0.4479	0.4479	456	512	512
Display Light Equivalent BT.709	DL White	0.7184	0.7184	0.7184	693	512	512
	DL Yellow	0.7185	0.7208	0.0311	651	203	539
	DL Cyan	0.0143	0.7239	0.7224	565	584	194
	DL Green	0.0652	0.7245	-0.0018	529	254	246
	DL Magenta	0.7389	-0.0027	0.7389	246	768	813
	DL Red	0.7424	0.0166	-0.0060	212	427	838
	DL Blue	0.0281	-0.0034	0.7591	115	850	494
Scene Light Equivalent BT.709	SL White	0.7500	0.7500	0.7500	720	512	512
	SL Yellow	0.7495	0.7501	0.0241	675	186	541
	SL Cyan	0.0188	0.7502	0.7499	584	586	184
	SL Green	0.0699	0.7511	0.0192	548	254	236
	SL Magenta	0.7510	-0.0081	0.7503	246	773	820
	SL Red	0.7514	0.0102	-0.0034	210	429	844
	SL Blue	0.0296	-0.0021	0.7514	115	846	495
Near Black Test Signal	-4%	-0.0383	-0.0383	-0.0383	30	512	512
	-2%	-0.0191	-0.0191	-0.0191	47	512	512
	-1%	-0.0095	-0.0095	-0.0095	55	512	512
	0%	0.0001	0.0001	0.0001	64	512	512
	1%	0.0096	0.0096	0.0096	72	512	512
	2%	0.0192	0.0192	0.0192	80	512	512
	4%	0.0383	0.0383	0.0383	97	512	512

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LUT 12-1 Details						
Variant	**LUT Type**	**LUT Cube Size**	**SDI LUT Mode Narrow: 64 - 940 (10-bit) Full: 0 – 1023 (10-bit)**	**Input Range [Min Input .. Max input]**	**Output Range [0.0 .. 1.0]**	**BT.709 colour matrix comp.**
12-1a	I	33	Narrow	Nominal	Nominal	No
12-1b	I	33	Narrow	Nominal	Nominal	Yes
12-1c	III	33	Full	Headroom	Headroom	No
12-1d	III	33	Full	Headroom	Headroom	Yes

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LUT 12-2: "Clean" (i.e. without graphics) feeds to match SDR BT.709 cameras – BT.2100 HLG to BT.709 down-mapping (scene-light)

Changes in v1.8: LUT headers have been updated with new licence information following open source release.

Deprecated conversion – few workflows require this conversion. Better results obtained with LUT 9.

For use with SDR cameras that implement a strict BT.709 OETF

BT.2100 HLG signals are down-mapped (tone-mapped) to BT.709, using a scene-light conversion. The "look" of the original BT.2100 HLG content is changed to match "native" BT.709 cameras, with a strict implementation of the BT.709 OETF.

This LUT is the *exact inverse* of LUT6-2 (BT.709 to BT.2100 HLG inverse tone-mapping, scene-light), so that the losses associated with 'round-tripping' (i.e. SDR-to-HDR-to-SDR conversion) are minimised.

"HDR Reference White" (75% HLG) is tone-mapped to 95% BT.709, allowing for some soft-clipping of highlights. 79% HLG signal is tone-mapped to 100% BT.709. With Type III LUTs that process sub-blacks and super-whites, the LUT outputs are clipped to EBU R.103 signal levels.

NOTE: Unlike BT.2100 HLG, SDR cameras that implement a strict BT.709 OETF tend to crush detail in the shadows of a scene. Shadow-detail visible in the HDR image will, as a result, become less evident after applying this conversion.

The 3D-LUT interpolation errors are greatest for the HDR to SDR down-mapping LUTs, as the output signals reach the extremities of the R′G′B′ signal range. For that reason, we provide the expected output values from a 33-cubed tetrahedral interpolator for the EBU Tech 3373 HLG Colour Bars, rather than the ideal values. Errors should be smaller with the 65-cubed LUTs and real pictures.

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	LUT 12-2 - BT.2100 HLG to BT.709, Down-Mapping (Scene-Light)						
	Test Input: BT.2100 HLG EBU Tech 3373	Expected BT.709 Output with tetrahedral interpolation					
		Nominal Range [0.0 ... 1.0]			10-Bit SDI		
		R'	G'	B'	Y'	C'_B	C'_R
100% HLG	100% White	1.0502	1.0502	1.0502	984	512	512
	100% Yellow	1.0502	1.0502	-0.0502	914	18	557
	100% Cyan	-0.0502	1.0502	1.0502	779	624	18
	100% Green	-0.0502	1.0502	-0.0502	709	131	64
	100% Magenta	1.0502	-0.0502	1.0502	294	892	959
	100% Red	1.0502	-0.0502	-0.0502	224	399	1005
	100% Blue	-0.0502	-0.0502	1.0502	89	1005	466
75% HLG	75% White	0.9463	0.9463	0.9463	892	512	512
	75% Yellow	0.9917	0.9617	-0.0502	848	55	567
	75% Cyan	-0.0502	1.0309	1.0005	763	609	28
	75% Green	-0.0502	1.0328	-0.0502	698	137	71
	75% Magenta	1.0502	-0.0502	1.0264	293	881	960
	75% Red	1.0502	-0.0502	-0.0502	224	399	1005
	75% Blue	-0.0502	-0.0377	1.0300	96	991	462
	40% Grey	0.4345	0.4345	0.4345	444	512	512
Display Light Equivalent BT.709	DL White	0.7170	0.7170	0.7170	692	512	512
	DL Yellow	0.7171	0.7194	0.0036	648	191	540
	DL Cyan	0.0024	0.7226	0.7211	562	585	189
	DL Green	0.0270	0.7233	0.0016	522	260	229
	DL Magenta	0.7380	-0.0014	0.7380	247	767	812
	DL Red	0.7415	0.0002	0.0014	202	436	844
	DL Blue	0.0005	0.0000	0.7586	112	851	481
Scene Light Equivalent BT.709	SL White	0.7492	0.7492	0.7492	720	512	512
	SL Yellow	0.7487	0.7494	0.0012	673	176	542
	SL Cyan	0.0077	0.7495	0.7491	582	587	179
	SL Green	0.0366	0.7504	0.0018	541	249	222
	SL Magenta	0.7503	0.0002	0.7496	251	770	817
	SL Red	0.7506	0.0003	0.0016	204	435	848
	SL Blue	0.0015	-0.0002	0.7507	111	848	481
Near Black Test Signal	-4%	-0.0076	-0.0076	-0.0076	57	512	512
	-2%	-0.0038	-0.0038	-0.0038	60	512	512
	-1%	-0.0019	-0.0019	-0.0019	62	512	512
	0%	0.0000	0.0000	0.0000	64	512	512
	1%	0.0019	0.0019	0.0019	65	512	512
	2%	0.0038	0.0038	0.0038	67	512	512
	4%	0.0076	0.0076	0.0076	70	512	512

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LUT 12-2 Details						
Variant	**LUT Type**	**LUT Cube Size**	**SDI LUT Mode Narrow: 64 - 940 (10-bit) Full: 0 – 1023 (10-bit)**	**Input Range [Min Input .. Max input]**	**Output Range [0.0 .. 1.0]**	**BT.709 colour matrix comp.**
12-2a	I	33	Narrow	Nominal	Nominal	No
12-2b	I	33	Narrow	Nominal	Nominal	Yes
12-2c	III	33	Full	Headroom	Headroom	No
12-2d	III	33	Full	Headroom	Headroom	Yes

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LUT 13: Test LUTs

Changes in v1.8: LUT headers have been updated with new licence information following open source release.

A set of four test LUTs with fixed outputs, intended for testing the configuration of 3D-LUT interpolators. See “LUTs for Testing Hardware” in the “Implementation Guidelines for HLG Format Conversion LUTs” for more details.

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LUT 14: Graded Content with P3D65 System Colour- PQ P3D65 1000 cd/m^2 to BT.2100 HLG

Changes in v1.8: LUT headers have been updated with new licence information following open source release.

Secondary conversion – conversions from P3D65 seldom required.

P3D65 PQ signals are converted to BT.2100 HLG in the 1000 cd/m^2 "bridge" condition, so that 1000 cd/m^2 PQ maps to 100% HLG. As such signals are only encountered in file-based workflows, Type II and Type III LUTs are not supplied.

The table illustrates the expected output signals from an ideal algorithmic converter, presented with a P3D65 PQ colour bar input. It takes no account of the expected interpolation or rounding errors.

LUT 14: PQ P3D65 Colour 1000 cd/m2 to BT.2100 HLG			
Test Input:	**Ideal BT.2100 HLG Output**		
PQ	**Nominal Range [0.0 ... 1.0]**		
Colour bars	**R'**	**G'**	**B'**
100% White	1.000	1.000	1.000
100% Yellow	1.000	1.000	0.573
100% Cyan	1.000	1.000	1.000
100% Green	1.000	1.000	0.599
100% Magenta	1.000	0.865	1.000
100% Red	1.000	0.829	0.000
100% Blue	0.870	0.603	1.000
58% White	0.749	0.749	0.749
58% Yellow	0.742	0.749	0.115
58% Cyan	0.450	0.748	0.757
58% Green	0.408	0.749	0.122
58% Magenta	0.744	0.236	0.783
58% Red	0.741	0.215	0.000
58% Blue	0.239	0.123	0.827
40% Grey	0.414	0.414	0.414
0% Black	0.000	0.000	0.000
−2% Black	0.000	0.000	0.000
+2% Black	0.013	0.013	0.013
+4% Black	0.025	0.025	0.025

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LUT 14 Details						
Variant	**LUT Type**	**LUT Cube Size**	**SDI LUT Mode Narrow: 64 - 940 (10-bit) Full: 0 – 1023 (10-bit)**	**Input Range [Min Input .. Max input]**	**Output Range [0.0 .. 1.0]**	**BT.709 colour matrix comp.**
14a	I	33	Narrow	Nominal	Nominal	No
14a65	I	65	Narrow	Nominal	Nominal	No

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LUT 15: BT.2100 HLG to PQ 1000 cd/m^2 with P3D65 System Colorimetry

Changes in v1.8: LUT headers have been updated with new licence information following open source release.

Secondary conversion – conversions to P3D65 seldom required.

BT.2100 HLG signals are converted to PQ signals with P3D65 colour in the 1000 cd/m^2 "bridge" condition, so that 100% HLG maps to 1000 cd/m^2 PQ. As such signals are only encountered in file-based workflows, Type II and Type III LUTs are not supplied.

The table illustrates the expected output signals from an ideal algorithmic converter, presented with an EBU Tech 3373 HLG colour bar input. It takes no account of the expected interpolation or rounding errors.

	LUT 15: BT.2100 HLG to PQ P3D65 Colour 1000 cd/m2 to			
	Test Input: BT.2100 HLG EBU Tech 3373	Ideal PQ P3D65 Output		
		Nominal Range [0.0 ... 1.0]		
		R'	G'	B'
100% HLG	100% White	0.7518	0.7518	0.7518
	100% Yellow	0.7570	0.7516	0.0000
	100% Cyan	0.0000	0.7521	0.7449
	100% Green	0.0000	0.7513	0.0000
	100% Magenta	0.7542	0.0000	0.7293
	100% Red	0.7549	0.0000	0.1928
	100% Blue	0.0000	0.0000	0.6922
75% HLG	75% White	0.5808	0.5808	0.5808
	75% Yellow	0.5857	0.5806	0.0000
	75% Cyan	0.0000	0.5810	0.5741
	75% Green	0.0000	0.5803	0.0000
	75% Magenta	0.5831	0.0000	0.5592
	75% Red	0.5837	0.0000	0.1120
	75% Blue	0.0000	0.0000	0.5243
	40% Grey	0.3916	0.3916	0.3916
Display Light Equivalent BT.709	DL White	0.4994	0.4994	0.4994
	DL Yellow	0.4982	0.4987	0.2840
	DL Cyan	0.3360	0.4930	0.4943
	DL Green	0.3350	0.4913	0.2652
	DL Magenta	0.4620	0.2015	0.4737
	DL Red	0.4575	0.1991	0.1611

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	LUT 15: BT.2100 HLG to PQ P3D65 Colour 1000 cd/m2 to			
	Test Input: BT.2100 HLG EBU Tech 3373	Ideal PQ P3D65 Output		
		Nominal Range [0.0 ... 1.0]		
		R'	G'	B'
	DL Blue	0.0000	0.0000	0.4517
Scene Light Equivalent BT.709	SL White	0.5097	0.5097	0.5097
	SL Yellow	0.5084	0.5082	0.2910
	SL Cyan	0.3449	0.5018	0.5032
	SL Green	0.3423	0.5001	0.2714
	SL Magenta	0.4660	0.2045	0.4775
	SL Red	0.4602	0.2012	0.1628
	SL Blue	0.0084	0.0000	0.4493
Near Black Test Signal	-4%	0.0000	0.0000	0.0000
	-2%	0.0000	0.0000	0.0000
	-1%	0.0000	0.0000	0.0000
	0%	0.0000	0.0000	0.0000
	1%	0.0123	0.0123	0.0123
	2%	0.0287	0.0287	0.0287
	4%	0.0610	0.0610	0.0610

LUT 15 Details						
Variant	**LUT Type**	**LUT Cube Size**	**SDI LUT Mode Narrow: 64 - 940 (10-bit) Full: 0 – 1023 (10-bit)**	**Input Range [Min Input .. Max input]**	**Output Range [0.0 .. 1.0]**	**BT.709 colour matrix comp.**
15a	I	33	Narrow	Nominal	Nominal	No
15a65	I	65	Narrow	Nominal	Nominal	No

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LUT 16: HDR Laser Projectors - BT.2100 HLG to X′Y′Z′ PQ at 108 cd/m^2

Changes in v1.8: LUT headers have been updated with new licence information following open source release.

Secondary conversion.

BT.2100 HLG signals are converted to PQ at a nominal peak luminance of 108 cd/m^2 using the non-linear X′Y′Z′ colour representation (D65 white point), suitable for HDR laser projectors. The EOTF gamma is calculated using the extended range gamma formula in Footnote 2 of ITU-R BT.2100, and a further adjustment made for dark viewing environments (see ITU-R Report BT.2390 Section 6.2).

LUT 16 Details						
Variant	**LUT Type**	**LUT Cube Size**	**SDI LUT Mode Narrow: 64 - 940 (10-bit) Full: 0 – 1023 (10-bit)**	**Input Range [Min Input .. Max input]**	**Output Range [0.0 .. 1.0]**	**BT.709 colour matrix comp.**
15a	I	33	Narrow	Nominal	Nominal	No
15a65	I	65	Narrow	Nominal	Nominal	No

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LUT17: SDR Graded Content – BT.2020 to BT.2100 HLG, Direct-Mapping (Display-Light)

Changes in v1.8: LUT headers have been updated with new licence information following open source release.

Secondary conversion – conversions from BT.2020 seldom required.

BT.2020 signals are directly-mapped into BT.2100 HLG at the BT.2408 signal levels using a display-light conversion. The "look" of the original BT.2020 content is therefore preserved on conversion. A 100% SDR signal is mapped to 75% HLG ("HDR Reference White").

The table below illustrates the output signals expected from an "ideal" converter, presented with 75% BT.2020 colour bars . The expected interpolation errors have not been factored in.

LUT 17: BT.2020 to BT.2100 HLG, Direct-Mapping (Display-Light)						
Test Input: 75% Colour bars	**Ideal BT.2100 HLG Output**					
	Nominal Range [0.0 ... 1.0]			**10-Bit SDI**		
	R'	**G'**	**B'**	**Y'**	**C'_B**	**C'_R**
75% White	0.6145	0.6145	0.6145	602	512	512
75% Yellow	0.6150	0.6150	0.0000	571	236	534
75% Cyan	0.0000	0.6173	0.6173	463	589	235
75% Green	0.0000	0.6180	0.0000	431	312	257
75% Magenta	0.6248	0.0000	0.6248	240	714	769
75% Red	0.6266	0.0000	0.0000	208	434	793
75% Blue	0.0000	0.0000	0.6400	97	799	489
40% Grey	0.3107	0.3107	0.3107	336	512	512
15% Grey	0.1004	0.1004	0.1004	152	512	512
0% Black	0.0000	0.0000	0.0000	64	512	512
–2% Black	-0.0103	-0.0103	-0.0103	55	512	512
+2% Black	0.0103	0.0103	0.0103	73	512	512
100% White	0.7500	0.7500	0.7500	721	512	512

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LUT 17 Details						
Variant	**LUT Type**	**LUT Cube Size**	**SDI LUT Mode Narrow: 64 - 940 (10-bit) Full: 0 – 1023 (10-bit)**	**Input Range [Min Input .. Max input]**	**Output Range [0.0 .. 1.0]**	**BT.709 colour matrix comp.**
17a	I	33	Narrow	Nominal	Nominal	No
17a65	I	65	Narrow	Nominal	Nominal	No
17b	I	33	Narrow	Nominal	Nominal	Yes
17c	III	33	Full	Headroom	Headroom	No
17d	III	33	Full	Headroom	Headroom	Yes

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LUT 18: SDR Graded Content – BT.2020 to BT.2100 HLG, Up-Mapping (Display-Light)

Changes in v1.8: LUT headers have been updated with new licence information following open source release.

Secondary conversion – conversions from BT.2020 seldom required.

BT.2020 signals are up-mapped (inverse tone-mapped) to BT.2100 HLG using a display-light conversion.

Whilst the ITU-R BT.2408 signal levels are taken into account, a modest boost is applied to the SDR highlights so that there is a closer match to natively-produced HDR content. The LUT does, however, attempt to preserve the artistic intent of the original BT.2020 content. 100% SDR signal is up-mapped to 82% HLG.

The table below illustrates the output signals expected from an "ideal" converter, presented with 75% BT.2020 colour bars. The expected interpolation errors have not been factored in.

LUT 18: BT.709 to BT.2100 HLG, Up-Mapping (Display-Light)						
Test Input: 75% Colour bars	**Ideal BT.2100 HLG Output**					
	Nominal Range [0.0 ... 1.0]			**10-Bit SDI**		
	R'	**G'**	**B'**	**Y'**	**C'_B**	**C'_R**
75% White	0.6144	0.6144	0.6144	602	512	512
75% Yellow	0.6150	0.6150	0.0000	571	236	534
75% Cyan	0.0000	0.6173	0.6173	463	589	235
75% Green	0.0000	0.6180	0.0000	431	312	257
75%	0.6248	0.0000	0.6248	240	714	769
75% Red	0.6266	0.0000	0.0000	208	434	793
75% Blue	0.0000	0.0000	0.6400	97	799	489
40% Grey	0.3107	0.3107	0.3107	336	512	512
15% Grey	0.1004	0.1004	0.1004	152	512	512
0% Black	0.0000	0.0000	0.0000	64	512	512
–2% Black	-0.0103	-0.0103	-0.0103	55	512	512
+2% Black	0.0103	0.0103	0.0103	73	512	512
100% White	0.8200	0.8200	0.8200	782	512	512

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LUT 18 Details						
Variant	**LUT Type**	**LUT Cube Size**	**SDI LUT Mode Narrow: 64 - 940 (10-bit) Full: 0 – 1023 (10-bit)**	**Input Range [Min Input .. Max input]**	**Output Range [0.0 .. 1.0]**	**BT.709 colour matrix comp.**
18a	I	33	Narrow	Nominal	Nominal	No
18a65	I	65	Narrow	Nominal	Nominal	No
18b	I	33	Narrow	Nominal	Nominal	Yes
18c	III	33	Full	Headroom	Headroom	No
18d	III	33	Full	Headroom	Headroom	Yes

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LUT19: HDR Cameras - BT.2100 Traditional Camera Look

Changes in v1.8: LUT headers have been updated with new licence information following open source release.

Secondary conversion – increasingly similar adjustments can be achieved using an HLG camera's painting controls.

HLG camera signals are converted to have a similar saturation and hue to those of a traditional BT.709 cameras. Use this LUT with care. It should only be applied to an HLG live programme output or camera, when the desire is to provide a close to the colour and saturation of an SDR BT.709 production covering the same live event. If more colourful images are required than those provided by the standard BT.2100 HLG "natural" look, it is usually better to use the saturation control of the camera's CCU rather than this LUT - the saturation control will usually avoid the hue shifts and saturation imbalance that's inherent within the BT.709 TV system.

The table illustrates the output signals expected from an "ideal" converter, presented with an EBU Tech 3373 HLG colour bar input. The expected interpolation errors have not been factored-in.

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	LUT 19 - BT.2100 HLG Traditional Camera Look						
	Test input: BT.2100 HLG EBU Tech 3373	**Ideal BT.2100 HLG Output**					
		Nominal Range [0.0 ... 1.0]			**10-Bit SDI**		
		R'	**G'**	**B'**	**Y'**	**C'$_B$**	**C'$_R$**
100% HLG	100% White	1.0000	1.0000	1.0000	940	512	512
	100% Yellow	1.0038	1.0022	0.3360	908	213	537
	100% Cyan	0.3626	1.0132	1.0092	802	592	221
	100% Green	0.3912	1.0158	0.3411	775	288	256
	100% Magenta	1.0482	0.4252	1.0335	611	707	769
	100% Red	1.0502	0.4305	0.2372	574	348	797
	100% Blue	0.3160	0.1815	1.0502	299	884	541
75% HLG	75% White	0.7500	0.7500	0.7500	721	512	512
	75% Yellow	0.7541	0.7523	0.1729	693	252	534
	75% Cyan	0.1867	0.7641	0.7599	600	582	253
	75% Green	0.2014	0.7669	0.1756	575	318	280
	75% Magenta	0.8012	0.2188	0.7857	419	693	752
	75% Red	0.8086	0.2216	0.1221	388	394	779
	75% Blue	0.1627	0.0934	0.8394	201	838	516
	40% Grey	0.3995	0.3995	0.3995	414	512	512
Display Light Equivalent BT.709	DL White	0.6142	0.6142	0.6142	602	512	512
	DL Yellow	0.6076	0.6158	0.2088	580	331	523
	DL Cyan	0.4011	0.6103	0.6215	551	543	418
	DL Green	0.3806	0.6115	0.1969	525	355	424
	DL Magenta	0.5847	0.2072	0.6509	355	664	665
	DL Red	0.5827	0.1977	0.0963	321	418	688
	DL Blue	0.1752	0.0897	0.7054	194	777	528
Scene Light Equivalent BT.709	SL White	0.6324	0.6324	0.6324	618	512	512
	SL Yellow	0.6260	0.6328	0.2172	595	327	524
	SL Cyan	0.4165	0.6261	0.6374	565	543	418
	SL Green	0.3948	0.6273	0.2042	538	352	423
	SL Magenta	0.5922	0.2107	0.6574	360	664	667
	SL Red	0.5880	0.2001	0.0975	323	417	689
	SL Blue	0.1734	0.0887	0.7011	193	776	528
Near Black Test Signal	-4%	-0.0365	-0.0365	-0.0365	32	512	512
	-2%	-0.0183	-0.0183	-0.0183	48	512	512
	-1%	-0.0091	-0.0091	-0.0091	56	512	512
	0%	0.0000	0.0000	0.0000	64	512	512
	1%	0.0091	0.0091	0.0091	72	512	512
	2%	0.0183	0.0183	0.0183	80	512	512
	4%	0.0365	0.0365	0.0365	96	512	512

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LUT 19 Details						
Variant	**LUT Type**	**LUT Cube Size**	**SDI LUT Mode Narrow: 64 - 940 (10-bit) Full: 0 – 1023 (10-bit)**	**Input Range [Min Input .. Max input]**	**Output Range [0.0 .. 1.0]**	**BT.709 colour matrix comp.**
19a	I	33	Narrow	Nominal	Nominal	No
19b	I	33	Narrow	Nominal	Nominal	Yes
19c	III	33	Full	Headroom	Headroom	No
19d	III	33	Full	Headroom	Headroom	Yes

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LUT20: SDR Graphics – sRGB to BT.2100 HLG, Direct-Mapping (Display-Light)

Changes in v1.8: LUT headers have been updated with new licence information following open source release.

Secondary conversion – in practice LUT 3 (BT.709 to BT.2100 HLG, Direct-Mapping) is usually sufficient.

sRGB (IEC 61966-2-1:1999) signals are directly-mapped into BT.2100 HLG at the BT.2408 signal levels using a display-light conversion. The "look" of the original sRGB content is therefore preserved on conversion. A 100% sRGB signal is mapped to 75% HLG ("HDR Reference White").

The table below illustrates the output signals expected from an "ideal" algorithmic converter, presented with a 75% sRGB colour bar input. The expected interpolation errors have not been factored in.

sRGB signals are only defined in the normalised range 0 to 1.0. So we provide Type I and Type II LUTs that do not process sub-black or super-white input signals.

LUT 20: sRGB to BT.2100 HLG, Direct Mapping (Display-Light)						
Test Input: SMPTE 219 75% Colour bars	**Ideal BT.2100 HLG Output**					
	Nominal Range [0.0 … 1.0]			**10-Bit SDI**		
	R'	**G'**	**B'**	**Y'**	**C'$_B$**	**C'$_R$**
75% White	0.6172	0.6172	0.6172	605	512	512
75% Yellow	0.6076	0.6147	0.2083	580	331	523
75% Cyan	0.3935	0.6016	0.6136	544	543	418
75% Green	0.3699	0.5989	0.1913	515	358	424
75% Magenta	0.5268	0.1829	0.5972	325	655	651
75% Red	0.5106	0.1695	0.0826	286	430	668
75% Blue	0.1343	0.0688	0.5934	167	739	522
40% Grey	0.3248	0.3248	0.3248	349	512	512
15% Grey	0.1246	0.1246	0.1246	173	512	512
0% Black	0.0000	0.0000	0.0000	64	512	512
−2% Black	N/A	N/A	N/A	N/A	N/A	N/A
+2% Black	0.0356	0.0356	0.0356	95	512	512
100% White	0.7500	0.7500	0.7500	721	512	512

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LUT 20 Details						
Variant	**LUT Type**	**LUT Cube Size**	**SDI LUT Mode Narrow: 64 - 940 (10-bit) Full: 0 – 1023 (10-bit)**	**Input Range [Min Input .. Max input]**	**Output Range [0.0 .. 1.0]**	**BT.709 colour matrix comp.**
20a	I	33	Narrow	Nominal	Nominal	No
20b	I	33	Narrow	Nominal	Nominal	Yes
20c	II	33	Full	Full/Nominal	Headroom	No
20d	II	33	Full	Full/Nominal	Headroom	Yes

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LUT21: On-set monitoring of live programmes – BT.2100 HLG to SDR BT.2020 (WCG) Down-Mapping (Display-Light)

Changes in v1.8: LUT headers have been updated with new licence information following open source release.

Secondary conversion – conversions to BT.2020 seldom required.

BT.2100 HLG signals are down-mapped (tone-mapped) to BT.2020 using a display-light conversion. Colours and the appearance of mid-tones and lowlights are maintained after conversion. The LUT, therefore, attempts to preserve the artistic intent of the original HLG content.

"HDR Reference White" (75% HLG) is mapped to 95% BT.709 and HDR highlights extend into the SDR super-white region; for that reason, only "Full-Range Mode" Type III LUTs are made available. 100% HLG signal is mapped to the EBU R103 preferred maximum of 105%. HLG signals above 100% are clipped.

This LUT is recommended for specialist applications such as on-set monitoring, where a high brightness wide colour gamut SDR display may be preferred over an HDR display.

The 3D-LUT interpolation errors are greatest for the HDR to SDR down-mapping LUTs, as the output signals reach the extremities of the R′G′B′ signal range. For that reason, we provide the expected output values from a 33-cubed tetrahedral interpolator for the EBU Tech 3373 HLG Colour Bars, rather than the ideal values. Errors should be smaller with the 65-cubed LUTs and real pictures.

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	LUT 21 - BT.2100 HLG to BT.2020, Down-Mapping (Display-Light)						
	Test Input: BT.2100 HLG EBU Tech 3373	Expected BT.2020 Output with tetrahedral interpolation					
		Nominal Range [0.0 ... 1.0]			10-Bit SDI		
		R'	G'	B'	Y'	C'_B	C'_R
100% HLG	100% White	1.0492	1.0491	1.0491	983	512	512
	100% Yellow	1.0191	1.0279	0.3800	929	223	532
	100% Cyan	0.3808	1.0044	0.9985	800	587	233
	100% Green	0.3390	0.9577	0.2997	726	295	259
	100% Magenta	1.0502	0.4681	1.0444	638	697	752
	100% Red	1.0469	0.3481	0.2644	525	387	828
	100% Blue	0.1695	0.2886	1.0229	328	856	432
75% HLG	75% White	0.9483	0.9483	0.9483	895	512	512
	75% Yellow	0.9339	0.9350	0.1389	841	156	540
	75% Cyan	0.1366	0.9341	0.9334	699	612	155
	75% Green	0.1421	0.9058	0.1232	641	257	198
	75% Magenta	0.9628	0.0586	0.9627	370	804	884
	75% Red	0.9559	0.0570	0.0420	320	393	915
	75% Blue	0.0122	0.0227	0.9455	130	927	474
	40% Grey	0.4968	0.4968	0.4968	499	512	512
Display Light Equivalent BT.709	DL White	0.7490	0.7490	0.7490	720	512	512
	DL Yellow	0.7353	0.7464	0.2926	692	310	523
	DL Cyan	0.4962	0.7285	0.7443	650	548	408
	DL Green	0.4713	0.7241	0.2725	617	342	415
	DL Magenta	0.6351	0.2615	0.7217	403	671	663
	DL Red	0.6169	0.2453	0.1344	359	416	682
	DL Blue	0.2016	0.1154	0.7143	216	770	529
Scene Light Equivalent BT.709	SL White	0.7774	0.7774	0.7774	745	512	512
	SL Yellow	0.7622	0.7728	0.3025	714	303	524
	SL Cyan	0.5137	0.7516	0.7687	668	549	405
	SL Green	0.4864	0.7472	0.2809	635	336	412
	SL Magenta	0.6442	0.2656	0.7315	408	673	664
	SL Red	0.6234	0.2482	0.1359	362	415	684
	SL Blue	0.2000	0.1137	0.7076	215	767	529
Near Black Test Signal	-4%	-0.0456	-0.0456	-0.0456	24	512	512
	-2%	-0.0228	-0.0228	-0.0228	44	512	512
	-1%	-0.0114	-0.0114	-0.0114	54	512	512
	0%	0.0000	0.0000	0.0000	64	512	512
	1%	0.0114	0.0114	0.0114	74	512	512
	2%	0.0227	0.0227	0.0227	84	512	512
	4%	0.0454	0.0454	0.0454	104	512	512

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LUT 21 Details						
Variant	**LUT Type**	**LUT Cube Size**	**SDI LUT Mode Narrow: 64 - 940 (10-bit) Full: 0 – 1023 (10-bit)**	**Input Range [Min Input .. Max input]**	**Output Range [0.0 .. 1.0]**	**BT.709 colour matrix comp.**
21c	III	33	Full	Headroom	Headroom	No
21d	III	33	Full	Headroom	Headroom	Yes

LUT 21 Header Metadata									
Variant	**LUT Type**	**Input**				**Output**			
		Colour Primaries	Matrix	Transfer Function	Extended Range Processing	Colour Primaries	Matrix	Transfer Function	Extended Range Processing
21c 21d	III	BT.2100 CICP 9	BT.2100 CICP 9	HLG CICP 18	1	BT.2020 CICP 9	BT.2020 CICP 9	BT.2020 CICP 14	1

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LUT22: BT.709 SDR Camera Shading – Combines LUT 4 followed by LUT 9 into a single LUT

Changes in v1.8: LUT headers have been updated with new licence information following open source release.

Primary conversion for live production.

BT.709 camera signals are first directly-mapped into BT.2100 HLG at the BT.2408 signal levels using a scene-light conversion, identical to LUT 4-1. They are then down-mapped back to BT.709 using the LUT 9 display-light conversion. This single LUT is therefore able to replicate the SDR signal delivered to SDR audiences, from an SDR camera used in a live HDR production. Depending on the production architecture, it may help reduce the number of converters required for shading SDR cameras. The "look" of the original BT.709 content is changed to match the subjective look of "native" BT.2100 HLG cameras. 100% SDR signal is directly-mapped to 75% HLG ("HDR Reference White"), which in turn is mapped back to 95% SDR.

The 3D-LUT interpolation errors are greatest for the HDR to SDR down-mapping LUTs, as the output signals reach the extremities of the R′G′B′ signal range. For that reason, we provide the expected output values from a 33-cubed tetrahedral interpolator for the EBU Tech 3373 HLG Colour Bars, rather than the ideal values. Errors should be smaller with the 65-cubed LUTs and real pictures.

LUT 22: SDR BT.709 Camera Shading LUT						
Test Input: SMPTE 219 75% Colour bars	**Expected BT.709 Output with tetrahedral interpolation**					
	Nominal Range [0.0 … 1.0]			**10-Bit SDI**		
	R'	**G'**	**B'**	**Y'**	**C'_B**	**C'_R**
75% White	0.7784	0.7784	0.7784	745.000	512.000	512.000
75% Yellow	0.7774	0.7774	0.0001	695.000	163.000	543.000
75% Cyan	0.0001	0.7750	0.7750	598.000	591.000	164.000
75% Green	0.0001	0.7737	0.0001	548.000	244.000	197.000
75% Magenta	0.7606	0.0001	0.7606	253.000	774.000	821.000
75% Red	0.7566	0.0001	0.0001	204.000	434.000	850.000
75% Blue	0.0001	0.0001	0.7412	110.000	844.000	481.000
40% Grey	0.4498	0.4498	0.4498	458.000	512.000	512.000
15% Grey	0.1714	0.1714	0.1714	214.000	512.000	512.000
0% Black	0.0001	0.0001	0.0001	64.0000	512.000	512.000
−2% Black	-0.0194	-0.0194	-0.0194	47.0000	512.000	512.000
+2% Black	0.0194	0.0194	0.0194	80.0000	512.000	512.000
100% White	0.9498	0.9498	0.9498	896.000	511.000	512.000

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LUT 22 Details						
Variant	**LUT Type**	**LUT Cube Size**	**SDI LUT Mode Narrow: 64 - 940 (10-bit) Full: 0 – 1023 (10-bit)**	**Input Range [Min Input .. Max input]**	**Output Range [0.0 .. 1.0]**	**BT.709 colour matrix comp.**
22c	II	33	Full	Full/Nominal	Headroom	No
22c65	II	65	Full	Full/Nominal	Headroom	No

LUT 22 Header Metadata									
Variant	**LUT Type**	**Input**				**Output**			
		Colour Primaries	Matrix	Transfer Function	Extended Range Processing	Colour Primaries	Matrix	Transfer Function	Extended Range Processing
22c 22c65	III	BT.709 CICP 1	BT.709 CICP 1	BT.709 CICP 1	1	BT.709 CICP 1	BT.709 CICP 1	BT.709 CICP 1	1

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